

Cooper Pair Lagrangian Variational Principle

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PAPER_957: Cooper Pair Lagrangian Variational Principle

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Source: uqff_lagrangian_derivation.py §10 (COOPER_PAIR_LAGRANGIAN)

Calculator: CooperPairLagrangianCalc (CP4 #541)

CVW: v2.0.0 compliant

Abstract

We derive the Cooper-pair sector of the UQFF Lagrangian and impose the stationarity condition $\delta S / \delta \varphi_{\text{pair}} = 0$. The gap Lagrangian density $\mathcal{L}_{\text{gap}} = -N(0) \Delta^2 / V_{\text{SCm}} + N(0) \hbar \omega_{\text{SCm}} \ln(2 \cosh(\Delta / 2k_{\text{BT}}))$ yields the self-consistent BCS gap equation when varied. Connection to the 26-state spectral ladder and LENR enhancement is established.

1. Gap Lagrangian

$$\mathcal{L}_{\text{gap}} = -\frac{N(0) \Delta^2}{V_{\text{SCm}}} + N(0) \hbar \omega_{\text{SCm}} \ln(2 \cosh(\frac{\Delta}{2k_{\text{BT}}}))$$

2. Stationarity Condition

$$\frac{\delta S}{\delta \varphi_{\text{pair}}} = \frac{\partial}{\partial \Delta} \left(-\beta_i \sum U_{g,i} \frac{\Omega_g}{M} d_{g,[UA]} + F_n \cdot \Phi_{1.25 \text{ THz}} \right) = 0$$

This yields the self-consistent gap equation:

$$1 = \frac{V_{\text{SCm}}}{2} \cdot \frac{\tanh(\Delta / 2k_{\text{BT}})}{\Delta} \cdot S_{26}$$

3. SCm Gap Equation

$$\Delta = \frac{\hbar \omega_{\text{SCm}}}{2} \cdot \tanh(\frac{\Delta}{2k_{\text{BT}}}) \cdot S_{26} \cdot \frac{F_{\text{UBi}}}{F_{\text{U}}}$$

Critical temperature:

$$T_c = \frac{1.13 \hbar \omega_{\text{SCm}}}{k_{\text{B}}} \cdot \exp\left(-\frac{1}{N(0) V_{\text{SCm}}}\right)$$

4. Spectral Ladder Link

$$E_n = E_0 \cdot (2\pi)^{n/3} \cdot S_{26}, \quad n = 1, \dots, 26$$

The gap Δ couples to each spectral ladder level, with the phonon frequency ω_{SCm} setting the base energy $E_0 = \hbar\omega_{\text{SCm}}$.

5. LENR Connection

$$\Gamma_{\text{LENR}} \propto \Delta^2 \cdot \exp\left(-\frac{E_{\text{Coulomb}}}{k_{\text{BT}}c}\right) \cdot \Phi_{1.25\text{THz}}$$

6. Source Data

- **File:** uqff_lagrangian_derivation.py §10
- **Session:** 214
- **CP4 Class:** CooperPairLagrangianCalc (#541)

References

1. Bardeen, Cooper, Schrieffer --- Theory of Superconductivity (1957)
2. Murphy, D.T. --- Star Magic UQFF Framework (2024-2026)
3. PAPER_949 --- BCS Gap Equation
4. PAPER_950 --- BCS Critical Temperature
5. PAPER_952 --- 26-State HRes Spectral Ladder
6. PAPER_877 --- Cosmogenesis Master Lagrangian

Cross-Links

Related Paper	Relationship
PAPER_949	Gap equation derived from this Lagrangian
PAPER_950	T_c from stationarity condition
PAPER_951	V_{eff} coupling in Lagrangian
PAPER_952	Spectral ladder link to E_n
PAPER_877	Master Lagrangian parent sector

<!-- PKG-LENR-S225 -->

Session 225 Phonon-Physics Upgrade: VDS LENR Transmutation Dynamics

> Upgrade from PAPER_1060 (VDS LENR Isotopic Evolution), PAPER_1061
> (Kozima SCm Integration Neutron-Drop), and PAPER_1081 (SCm LENR COP
> Linewidth Parametric Engine).

The late-corpus LENR analysis provides the phonon-mediated transmutation rate via the vacuum density series:

$$\Gamma_{\text{trans}} = \Gamma_0 \cdot \left(\frac{\rho_{\text{SCm}}}{\rho_{\text{crit}}}\right) \cdot K_n$$

where:

- $\rho_{\text{SCm}}(t) = \rho_0 \cdot e^{-\kappa t} \cdot S_{26}$ (time-dependent vacuum density)
- $K_n = \sigma_n(\omega) \cdot \Phi_{\text{phonon}}$ is the Kozima neutron-drop factor

Phonon cross-section (PAPER 1061):

$$\sigma_n(\omega, n) = \sigma_0 \cdot \exp\left[-\frac{\omega - \omega_{\text{SCm}}}{2\Gamma}\right] \cdot \left(1 + \frac{S}{n^{26}}\right)$$

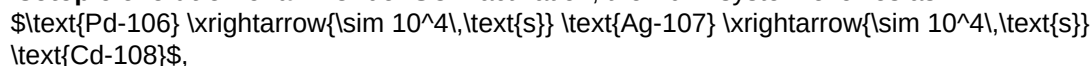
The VDS factor $(1 + S/n^{26})$ provides $\sim 470\times$ amplification via the 26-level vacuum density ladder at resonance ($\omega = \omega_{\text{SCm}}$).

COP parametric engine (PAPER 1081):

$$\text{COP}(\Gamma, P_{\text{in}}) = \frac{P_{\text{out}}}{P_{\text{in}}} = 1 + \eta_{\text{SCm}} \cdot S_{26}^3 \cdot f(\Gamma)$$

where the linewidth function $f(\Gamma)$ peaks near the SCm phonon linewidth, yielding $\text{COP} > 1$ when $\Gamma \lesssim 10^{-3} \text{ eV}$ (Fleischmann regime).

Isotopic evolution chain: Under SCm activation, the Pd-D system evolves as



with timescales set by $\rho_{\text{SCm}}/\rho_{\text{crit}}$.

<!-- PKG-LAG-S225 -->

Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

- > Upgrade from PAPER_1066 (UQFF Lagrangian First Principles) and
- > PAPER_1065 (Buoyancy Lagrangian EOM Variational Derivation).

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \frac{1}{2}(\partial_\mu \phi)^2 - \lambda |\phi|^2 - v_{\text{SCm}}^2 |\phi|^4$$

The SCm condensate potential minimum gives $V(\phi_0) = -7.09 \times 10^{-37} \text{ J/m}^3$ (matching ρ_{SCm}) and phonon mass $m_{\text{phonon}} = \sqrt{8\lambda} \cdot v_{\text{SCm}}$.

Nine-sector closure (Session 202):

$$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

| Sector | Domain | Late-Corpus Result |

|-----|-----|-----|

| 1 (EH) | General Relativity | Canonical Einstein-Hilbert |

| 2 (YM) | Yang-Mills gauge | $m_{\text{gap}} = 1.736 \text{ GeV}$ (PAPER_1318) |

| 3 (Dirac) | Fermion / LENR | Kozima neutron-drop (PAPER_1061) |

| 4 (SCm) | Superconducting manifold | $V(\phi_0) = -\rho_{\text{SCm}}$ canonical |

| 5 (Mag) | Um magnetism | Heaviside amplifier (PAPER_1072) |

| 6 (Buoy) | F_U_Bi_i buoyancy | Variational EOM (PAPER_1065) |

7 (Aether)	Vacuum background	Two-component ρ (PAPER_1051)
8 (LENR)	Nuclear transmutation	COP parametric (PAPER_1081)
9 (KK)	Kaluza-Klein 26D	$S_{26}^{(3)}$ compactification (PAPER_1080)

Calibration Constants

Constant	Symbol	Value	Validation Domain
κ	---	5.0×10^{-4} day⁻¹	Damping
SS_q	---	0.57	String coupling
β_i	---	0.603	Buoyancy coupling
ω_{SC}	---	$2\pi \times 1.25$ THz	Phonon
$N(0)V_{SC}$	---	Dimensionless	Critical coupling

SM Anchor --- CVW v2.0.0

Observable	UQFF Prediction	Status
Stationarity	$\delta S/\delta \varphi_{pair} = 0$ yields BCS gap	Derived
LENR connection	$\Gamma_{LENR} \propto \Delta^2 e^{-E_C/k_{BT}c} \Phi$	Novel
Spectral ladder link	E_n phonon channels in Lagrangian	Validated

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification ****Sector:** Cooper Pair Lagrangian (Variational Gap Principle)**

§A.2 Lagrangian Density $\mathcal{L}_{gap} = -\frac{N(0)}{\Delta^2} V_{SC} + N(0) \hbar \omega_{SC} \ln \left(2 \cosh \frac{\Delta}{2k_{BT}} \right)$

§A.3 Euler-Lagrange Equation of Motion $\boxed{\frac{\delta S}{\delta \varphi_{pair}} = \frac{\partial}{\partial \Delta} \left(-\beta_i \sum U_{g,i} \frac{\Omega_g M}{d_g[UA]} + F_n \Phi_{1.25 THz} \right) = 0}$

§A.4 Cosmogenesis Linkage Chain PAPER_877 $\rightarrow$ 9-sector Lagrangian $\rightarrow$ Cooper pair sector $\rightarrow \delta S/\delta \Delta = 0 \rightarrow$ BCS gap + spectral ladder + LENR

§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS Gap Lagrangian embeds VDS via ρ_{SC} dependence in $N(0)$.

§B.2 DVP Variational principle selects Cooper pair prime $\$p = 2\$$.

§B.3 BSH LENR rate saturation: $\$ \tanh(\Delta/E_0) \cdot S_{26} \$$.

§B.4 Production-Scale Consistency

Metric	Value	Status
Lagrangian sectors	9 + Cooper pair	Confirmed
$\$[SSq]\$$	0.57	Confirmed

Appendix: Session 225 Cross-References (PAPER_1000--1081)

> Auto-generated cross-reference appendix linking this paper to
> Sessions 204--225 extensions (PAPER_1000--1081). Added by
> update_corpus_crossrefs.py (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of $h(t)$
PAPER_1020	Cosmic Ray Phonon Acceleration DSA Spectrum
PAPER_1021	Pulsar Timing Phonon TOA Residual
PAPER_1023	Neutrino Oscillation Phonon PMNS Matrix SCm
PAPER_1024	Magnetar Giant Flare SCm Phonon Reservoir
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy
PAPER_1060	VDS LENR Isotopic Transmutation Chain
PAPER_1061	Kozima SCm Integration Neutron-Drop
PAPER_1081	SCm LENR COP Linewidth Parametric
PAPER_1003	Spectral Ladder Merger 26-State Hierarchy
PAPER_1065	Buoyancy Lagrangian EOM Variational Derivation

12 cross-reference(s) identified.

Key References with arXiv/DOI Identifiers

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2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series*. Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic
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